DHAYALAN K

Frontend Engineer | React, TypeScript, Modern CSS/Tailwind & Responsive Web Applications

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IMPACT AT A GLANCE

Client-Side E-Commerce Architecture shipped Shahihennaholic with instant LocalStorage cart persistence | **Modular Single-Page UI Extension** FLARUM1 written in JavaScript hooking into Mithril.js component lifecycles | **Sub-Second First Contentful Paint** mobile-first responsive design, zero layout shift & optimized assets | **Real-Time Browser DOM Telemetry** session instrumentation tool capturing DOM mutations and event streams

Detail-driven Frontend Engineer focused on building responsive, performant, and accessible user interfaces. Skilled in React (v18+), modern JavaScript (ES6+), TypeScript, Tailwind CSS, and single-page application lifecycles. Experienced in engineering real-world e-commerce experiences with client-side state persistence (LocalStorage), authoring modular UI extension packages hooking into modern component runtimes, and building real-time browser event telemetry tools. Passionate about design systems, responsive typography, and fast load speeds.

FRONTEND SYSTEMS & COMPONENT ARCHITECTURE

E-Commerce Web Architecture & Client State

- Engineered **Shahihennaholic Bridal E-Commerce Storefront**: responsive web catalog featuring instant client-side shopping cart calculation and persistence without backend dependencies.
- Designed mobile-first UI with fluid product image galleries, instant page responsiveness, zero cumulative layout shift, and automated WhatsApp checkout payload serialization.
- Developed **FLARUM1**: independent frontend extension package extending the Flarum community platform with customized UI styling, themes, and interactive hooks.
- Programmed lifecycle hooks into single-page application router events, modifying DOM components dynamically without modifying core upstream source code.

Telemetry & Interactive Dashboard Engineering

- Created **Browser DOM Telemetry & Interaction Monitor**: captured active DOM mutations, click paths, and scroll interactions in real-time with zero-jank JSON buffer batching.
- Built **Interactive DevOps & Systems Dashboard**: dark/light theme systems dashboard featuring responsive multi-node telemetry grids, particle background animations, and dynamic wire protocol bitmask toggles.

TECHNICAL SKILLS

Core Frontend: JavaScript (ES6+ / ESNext), HTML5 Semantic Architecture, CSS3 (Modern Flexbox & CSS Grid), TypeScript, DOM API, Web Vitals

Frameworks & UI Libraries: React (v18+), Functional Components, Custom Hooks, Context API, Next.js (App Router), Tailwind CSS, Mithril.js, Responsive Mobile-First Design

State & Storage: React State & Reducers, Context API, Browser LocalStorage, SessionStorage, Event Queues, JSON Buffer Serialization

Build Tools & Performance: Vite, Webpack, PostCSS, Responsive Images, Git/GitHub, Chrome DevTools Profiling, Lighthouse Audit, Postman

Design-to-Code & Standards: Design Systems, WCAG Accessibility, Cross-Browser Compatibility, Micro-Interactions, Component Lifecycle

PROFESSIONAL EXPERIENCE

Frontend Web Developer, Web Platforms & Client Projects **2024 to Present, Coimbatore**

- Built and styled e-commerce storefronts with mobile-first UI, fluid product galleries, and client-side cart storage.
- Customized community forum platform with modular JavaScript extensions (FLARUM1) hooking into SPA component lifecycles.
- Optimized frontend asset loading and CSS architecture, achieving sub-second initial render times.

Systems & Telemetry Engineer, Independent Systems Lab **2025 to Present, Coimbatore**

- Built Python telemetry tools to monitor live browser DOM event streams and capture real-time performance queues.
- Developed responsive web dashboard with dark/light themes and canvas animations without heavy external frameworks.

KEY PROJECTS & CODEBASES

Shahihennaholic Bridal E-Commerce Storefront github.com/DHAYALANKANIAPPAN/PROJECTS

- Engineered responsive web catalog with interactive shopping cart and LocalStorage state persistence.
- Built automated WhatsApp order payload generator formatting line items and customer details. Stack: HTML5, CSS3, JavaScript.

FLARUM1 — Modular UI Component Extension github.com/DHAYALANKANIAPPAN/flarum

- Authored frontend extension hooking into single-page application router events for dynamic styling without core modifications.
- Constructed modular CSS sheets with custom responsive breakpoints. Stack: JavaScript, Mithril.js, CSS3.

Browser DOM Telemetry & Interaction Monitor github.com/DHAYALANKANIAPPAN/PROJECTS

- Constructed telemetry utility capturing DOM mutations and user interactions in real-time with zero-jank buffer serialization.

EDUCATION

Bachelor of Engineering in Computer Science and Engineering (Cyber Security), Pursuing

Sri Shakthi Institute of Engineering and Technology, Coimbatore (2024 – 2028) | Grade Point Average: 8.21 out of 10.0.
Available for full-time employment & internships.

VERIFIED CERTIFICATIONS, AWARDS & ACTIVITIES

- Certificate of Merit: Java + Mongo DB (92% Score) — Hazhtech Career Development Centre / IACT (Programming fundamentals).
- Amazon ECS Knowledge Badge Assessment — Amazon Web Services (AWS) · Credly Official.
- Cloud Game Development Knowledge Badge — Amazon Web Services (AWS) · Credly Official.
- Yukthi CTF 2.0 Participant — National-Level Capture The Flag competition, Tamil Nadu Police.
- 0xTi - HackTiVate 24-Hour CTF Participant — Rajalakshmi Engineering College, Chennai.